

AK VisionFlow — Phase 4

Claude Development Prompt
Production Hardening + Advanced Business Command Center

Critical context: Phase 1, Phase 2 and Phase 3 are already implemented. This phase is an incremental upgrade. Inspect the existing repository first. Do not rebuild working functionality, reset the database, or delete/corrupt existing tenant data.

1. Mission

Bring the existing AK VisionFlow application to production-grade readiness and implement the Advanced Admin/Owner Business Command Center. Preserve the current multi-tenant architecture, POS, inventory, purchases, optical orders, prescriptions, payments, RBAC, offline engine, cloud sync, PWA foundation and existing tests.

2. Product Rules

- AK VisionFlow is generic SaaS for optical shops, eye-care businesses and medical stores; never hard-code one organization.
- Existing tenants, users, sales, purchases, inventory, invoices, payments, prescriptions and optical orders must survive the update.
- Application update must never mean database reset.
- All schema changes must use safe, versioned migrations.
- Preserve tenant isolation, backend-authoritative financial calculations and reversal/soft-delete principles.
- Preserve offline-first operation, queue/retry behavior and duplicate prevention.
- Do not introduce breaking API changes without compatibility/migration.
- Run existing tests before and after implementation; add regression tests for new work.

3. Production Hardening

- Verify live frontend and backend deployments, health endpoints, environment variables and API connectivity.
- Verify production CORS configuration and unauthorized-origin rejection.
- Verify Neon/PostgreSQL schema consistency, migrations and backup/PITR configuration.
- Perform or document a safe non-production restore test and recovery procedure.
- Improve operational logging for critical errors and sync failures.
- Document rollback and database recovery procedures.
- Check performance of dashboard/report queries and avoid obvious N+1 or inefficient queries.
- Regression-test PWA installation, offline transactions, sync retries and recovery.
- Re-verify important reports and critical sales, purchase, inventory, payment and reversal workflows.
- Review dependencies for known vulnerabilities and document any unresolved non-critical findings.

4. Advanced Admin / Owner Business Command Center

Create a premium single-screen dashboard that consolidates the owner's business information. It must complement, not unnecessarily replace, the existing basic dashboard.

KPI cards: Today's Sales, Gross Profit, Net Profit where accounting data exists, Purchases, Expenses, Cash, Bank, Receivables, Payables, Inventory Value.

Operational widgets: sales trend, profit trend, top-selling products, most profitable products, low stock, slow-moving stock, dead stock, expiring medicines, pending optical jobs, delayed orders, ready orders, new vs returning customers, outstanding payments, branch performance and staff performance.

Filters: today/yesterday/week/month/custom date range, branch, business area/module, category, product, supplier, customer, staff/user, payment status and order/job status.

Customization: authorized users can show/hide widgets, reorder them where practical, save dashboard preferences and use sensible default layouts. Preferences must remain tenant/user scoped.

Drill-down: every meaningful KPI/widget should open the relevant filtered records/report. Examples: Low Stock → Inventory; Receivables → Customer Ledger; Pending Jobs → Optical Jobs; Branch Sales → Branch Report.

5. Dashboard UX

- Professional medical/optical SaaS interface; responsive on desktop, tablet and mobile/PWA.
- Fast initial rendering using efficient aggregation/lazy loading where appropriate.
- Clear loading, empty and error states.
- Show sync/last-updated information where useful.
- Respect existing light/dark mode.
- Respect RBAC, tenant scope and branch scope.
- Never expose unauthorized financial or customer information.

6. Security & Data Integrity

- Re-test cross-tenant access for all tenant-owned resources and foreign-key ownership checks.
- Verify RBAC for sensitive operations.
- Verify audit logging for critical administrative and financial actions.
- Verify server-side validation and authorization.
- Verify client input cannot override backend financial totals.
- Verify reversals/voids do not create phantom balances or stock.
- Verify no secrets are present in source code, logs or client bundles.

7. Testing & Acceptance

- All existing automated tests pass.
- Add tests for dashboard authorization, tenant isolation, filters, date ranges and aggregation accuracy.
- Test empty datasets and realistic larger datasets.
- Regression-test sales, purchases, inventory, optical orders, payments, reversals and sync.
- Verify no duplicate records or sync operations are introduced.
- Verify existing data remains readable and usable after deployment.
- No critical security, data-integrity or production-blocking issue remains open.

8. Required Deliverables

- Production-hardening fixes and verification.
- Advanced Admin/Owner Business Command Center.
- Dashboard filters, widgets, customization and drill-downs.
- Backup/recovery verification and documented procedure.

- Production deployment/configuration checklist.
- Security and regression test results.
- Database migrations if required.
- Updated technical documentation.
- Final Phase 4 report: implemented items, files changed, migrations, tests, known limitations and next steps.

9. Claude Execution Order

1. Inspect repository and establish the Phase 1–3 baseline.
2. Run the existing test suite and record the baseline.
3. Verify production deployment/configuration and database readiness.
4. Complete security, tenant-isolation and data-integrity checks.
5. Complete backup/recovery and operational readiness checks.
6. Implement the Advanced Business Command Center.
7. Implement filters, widgets, customization and drill-downs.
8. Add automated regression tests.
9. Run the complete test suite again.
10. Perform final production-readiness review and document results.

10. Final Instruction to Claude

Build Phase 4 as a safe incremental upgrade to the existing AK VisionFlow application. Preserve everything that already works. Do not rebuild the core. Do not reset the database. Do not use destructive migrations. Prioritize production reliability, data safety, tenant isolation, backup/recovery and the Advanced Admin/Owner Business Command Center. The result must be tested, responsive, maintainable and ready for controlled real-world deployment, while providing a stable foundation for Phase 5: Complete Accounting + Professional Procurement.